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Overview

Since 2007, African swine fever (ASF) has spread through Europe, Asia and some of Australia's nearest Pacific neighbours. The disease affects both domestic and feral (wild) pigs and has no approved vaccines or effective treatments. ASF is responsible for significant economic losses in affected countries. The potential costs of this key threat entering Australia include the costs of eradication, costs to industry until the disease is eradicated, and long-term costs of ASF's presence should eradication prove economically or technically infeasible.

ABARES estimated the costs to the economy of 2 plausible scenarios for an ASF incursion into Australia - incursions into either the feral pig population or the domestic production system - as well as the costs of ASF becoming endemic.

These estimates provide insights into the value of keeping ASF out of Australia. Should an incursion occur in the future, the endemic cost could be used to assess the costs and benefits of eradication decisions.

Key findings

- A small-scale outbreak of ASF in domestic pigs followed by eradication of the disease was estimated to cost \$117 million to \$263 million.
- A small-scale outbreak of ASF in feral pigs followed by eradication of the disease was estimated to cost \$101 million to \$127 million.
- Endemic ASF was estimated to cost between about \$0.4 billion and \$2.5 billion.

Characteristics of African swine fever (ASF) virus and the epidemiology of ASF that have implications for the form of an eradication response, the way in which endemic disease might be managed, and the consequent costs of both, include:

- High virulence – we assumed outbreaks are caused by ASFV genotype II. The disease will progress rapidly in infected pigs and mortality is close to 100%. This also contributes to slow natural spread in wild pig populations overseas.
- Long persistence in the environment – the virus can remain viable for extended periods in some substrates. Contaminated meat, feed, equipment, clothing, and infected carcasses pose a high risk for disease transmission.
- Low prevalence – the virus is rarely detected in healthy pigs, so surveillance and testing must focus on sick or dead pigs.
- Non-zoonotic – the virus poses no risk to human health. Meat from pigs remains safe for human consumption.

A small-scale outbreak of ASF in domestic pigs followed by eradication of the disease was estimated to cost \$117 million to \$263 million over 5 years. Key features affecting this estimate were:

- Whether exports of products are impacted according to current certification requirements or additional restrictions occur.
- The consumer response to the outbreak - Although ASF poses no risk to human health, perceived risks from pork consumption may reduce Australian consumer demand. The plausible high end of reduced demand for pork during this outbreak scenario added more than \$100 million to the estimated cost of the outbreak. This suggests that there is high value in activities that would maintain confidence and reduce negative perceptions that consumers might have.
- Requirements for destruction and disposal at the infected abattoir and the length of time to resume movements of pigs from trace premises to slaughter. These response decisions were estimated to cost between \$34 million and \$59 million. Response planning would ideally seek to minimise these costs while maximising the likelihood of eradication through attention to whole-of-supply chain biosecurity.

Key features of the Australian pig industry with implications for the costs of ASF and the response to an incursion

Several features of the Australian pig industry are likely to influence the costs associated with the disease and response to an incursion. These include:

- No redundancy – any delay in pig movements will result in farms holding more pigs than they can accommodate, leading to animal welfare problems developing.
- Few abattoirs, largely operating at capacity – any reduction in capacity could result in long-distance interstate movements of pigs to another facility (if capacity, logistical, and welfare considerations allow) or backlogs of slaughter-ready pigs on-farm, exacerbating animal welfare issues.
- Many interstate movements – restrictions on interjurisdictional movements during a disease response will cause major disruptions to normal pig flows within the system.
- Most offal is exported – as there is almost no domestic market for offal, excess will need to be disposed of resulting in abattoirs losing revenue from offal sales and incurring disposal costs – this reduction in profitability is likely to be reflected in higher slaughter fees.
- A small number of producers own most of the herd – there are many small enterprises, which tend to have poorer on-farm biosecurity measures (Schembri et al. 2015).

A small-scale outbreak of ASF in feral pigs followed by eradication of the disease was estimated to cost \$101 million to \$127 million over 5 years. To eradicate would require:

- Feral pig population reduction measures.
- Search for, and collection and disposal of, contaminated carcasses and substrates.
- Surveillance for delimitation and proof of freedom.

The biggest contributor to the estimated cost was the cost of human resources required to carry out the eradication campaign. Market costs were also substantial and varied depending on whether exports of products are impacted according to current certification requirements or additional restrictions occur.

Endemic ASF in the feral pig population will have negative impacts but could also have benefits.

For the pig meat industry negative impacts come from:

- the increased costs of production that arise from enhanced on-farm and supply chain biosecurity practices
- temporary closures of export markets due to the initial incursion, which last longer than in eradication scenarios occasional outbreaks in the production
- system due to spill over of disease from the feral population.

Costs are also likely to be incurred for feral pig control and ongoing communication activities.

Costs of these negative impacts range from \$0.82 billion to \$2.8 billion. These estimates do not include negative social impacts, which are also likely.

Potential benefits of endemic ASF in the feral pig population include a reduction in the environmental impacts of feral pigs and production gains from other industries affected negatively by feral pigs.

Estimates of other industry benefits ranged between \$88 million and \$1 billion.

Considerable uncertainty exists around the cost of endemic ASF. The speed of disease spread, which we assumed will correlate with the implementation of enhanced on-farm and supply chain biosecurity practices, and the frequency of spill over outbreaks in the domestic production system are key drivers.

Estimated costs of endemic ASF

Cost component	Cost estimate (\$2018–19 million)a			
	Slow spread		Rapid spread	
	Low estimate	High estimate	Low estimate	High estimate
Market closures	722.8	821.2	1,671.6	1,770.1
Response – feral pig management	40.0	43.0	51.9	54.9
Response – enhanced biosecurity	85.5	288.0	287.0	430.8
Response – response to outbreaks in domestic pigs	0	490.0	0	490.0
Response – communications and export market negotiations	44.4	44.4	50.6	50.6
Benefits	-453.9	-88.4	-1,040.8	-256.4
Total	438.8	1,598.3	1,020.4	2,540.0

Note. All figures are derived using a discount rate of 3%.

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